

Assignment 1

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Subject

Programming Fundamentals

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Question No 1

Source Code^(a):

Source code is human readable program written by a programmer in high level language. It is understandable by humans and must be compile into machine code before its execution.

Example:

Machine Code: Machine code is in machine language (binary 0 and 1) that computer can understand easily. Generated from source code through compiler.

Example: 10101001 10110001

Java Convert Source code into machine code:

Java convert source code into machine code in following steps:-

Source Code → compile → byte code → Link → Execute/Run

First source code is compiled into byte code by compiler, which links with linker code and executes or runs.

(b)

Logical Errors: Logical errors happen when the program runs but gives wrong output because of wrong logic.

Example:

```
int a = 5;      int b = 10;  
int sum = a - b    // wrong logic it should be (a+b)  
System.out.println(sum);
```

Syntax Errors: Happens when you break the grammar rules of Java. The program will not compile.

Example:

```
int x = 5    // Missing (;) is syntax error.
```

Specify Errors in Java program

1)

`System.out.print("test");` (Syntax Error)
Out → out Wrong quotation marks → "

`System.out.print("test");` // True

integer price;

(int keyword)

`int price;` // True

Specify Errors:

`float tax of product = price * 0.06;`

(Syntax Error)

Variable contain no spaces tax of product
Price not declared (assumed)

`float taxOfProduct = price * 0.06f // True`

keyWords: (C)
keywords are reserved words in Java that have special meaning in Programming language and not used as Identifiers.

Example:

int, float, if, void etc.

Identifiers: Identifiers are the name given to the variables, classes and methods. We not use keywords as identifier.

Example:

int marks;

Marks → Identifier

Operators: Symbols used to perform operations called operators.

Example:

Operators

Arithmetic:

+ , - , * , / , %

Logical operators:

|| , &&

Relational:

> , < , ==

Question No 2

Write pseudocode of programs.

(a)
Read the radius and length
to calculate Area & Volume.

Read radius
Read length
 $\text{area} = \text{radius} * \text{radius} * 3.1416$
 $\text{Volume} = \text{area} * \text{length}$
Print area
Print volume

```
11 import java.util.Scanner;
12 public class ArithmeticCalculations {
13
14     /**
15      * @param args the command line arguments
16      */
17     public static void main(String[] args) {
18         System.out.println("enter radius");
19         Scanner input=new Scanner(System.in);
20         double radius=input.nextDouble();
21         System.out.println("enter length");
22         double length=input.nextDouble();
23         double area=radius*radius*length;
24         double volume=area*length;
25         system.out.println("area is="+area+"\n"+"volume is="+volume);
26     }
27 }
```

arithmeticcalculations.ArithmeticCalculations > main >

Output - ArithmeticCalculations (run) X

```
run:
enter radius
5.5
enter length
5.7
area is=172.425
volume is=982.8225000000001
BUILD SUCCESSFUL (total time: 8 seconds)
```

(b)
Get Price from user , use tax
rate to calculate amount
of tax. 17% Tax.

Start

Read price

$\text{tax} = \text{price} \times 0.17$

$\text{total} = \text{price} + \text{tax}$

Print tax

Print total

End

```
1  import java.util.Scanner;
2  public class TaxCalculate {
3
4      /**
5       * @param args the command line arguments
6       */
7      public static void main(String[] args) {
8
9          System.out.println("enter price");
10         Scanner sc=new Scanner(System.in);
11         double price=sc.nextDouble();
12         double tax=price*0.17;
13         double totalPrice=price+tax;
14         System.out.println("total amount="+totalPrice);
```

Output - TaxCalculate (run) X

run:

enter price

34.5

total amount=40.365

BUILD SUCCESSFUL (total time: 17 seconds)

Trip cost program: (C)

Start
Read distance
Read milePerGallon
 $\text{Gallons} = \text{distance} / \text{milesPerGallon}$
 $\text{cost} = \text{gallons} * \text{PricePerGallon}$
Print cost
End.

```
1 import java.util.Scanner;
2 public class TripCalculation {
3
4     /**
5      * @param args the command line arguments
6      */
7     public static void main(String[] args) {
8         System.out.println("Enter distance");
9         Scanner sc=new Scanner(System.in);
10        double distance =sc.nextDouble();
11        System.out.println("enter miles per gallon ");
12        double milesPerGallon=sc.nextDouble();
13        double gallonUsed =distance/milesPerGallon;
14        System.out.println("Enter price per Gallon");
15        double pricePerGallon=sc.nextDouble();
16        double cost=pricePerGallon*gallonUsed;
17        System.out.println("total cost for trip =" +cost);
18    }
19 }
```

tripcalculation.TripCalculation >

Output - TripCalculation (run) X



Enter distance

23.4



enter miles per gallon

45.6



Enter price per Gallon



34.7

total cost for trip =17.80657894736842

BUILD SUCCESSFUL (total time: 10 seconds)

Question No 3

```
0  */
1  public class Assignment_1 {
2
3  /**
4   * @param args the command line arguments
5   */
6  public static void main(String[] args) {
7      int x=2;
8      int y=5;
9      int z=3;
10     System.out.println(x*y/x);
11     System.out.println(x*(y/x));
12     System.out.println(y*(x/y+x/y));
13     System.out.println(y*x/y+y*x/y);
14     System.out.println(3*-2%4);
15 }
```

assignment_1.Assignment_1 > main >

Output - Assignment_1 (run) X

run:

5

4

0

4

-2

BUILD SUCCESSFUL (total time: 0 seconds)